

Anurag Aggarwal

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EDUCATION

University of Western Ontario

London, Ontario

BESc in Mechatronics and Artificial intelligence systems Engineering (Dual Degree) - Cum. GPA: 3.9

Sept. 2023 - Current

Relevant Courses: Algorithms & Data Structures, Digital Logic Systems and Design, Computer Architecture, Cyber physical Systems, Electrical Instrumentation, Software design for Sys Eng.

WORK EXPERIENCE

Tethos Association

London, Ontario

Embedded Systems and Mechatronics Developer

Sept. 2025 - Present

- Collaborated in material selection, **glove casing design and circuit prototypes** with mechanical and electrical teams.
- Programmed **Arduino Uno** to integrate **gyroscope, flex sensors**, etc. improving input reliability for the interactive game.
- Provided **hardware - software integration** guidance to align sensor data with game logic, reducing iteration and rework.
- Helped enable accurate **glove-to-hand motion mapping**, improving responsiveness and realism of the control interface.

Kissan Engineers (Casting)

Batala, Punjab, India

Computer Vision & AI-Powered Automation Intern

May - Sept. 2025

- Designed a **CV-based defect-inspection ML model** for identifying improperly cast parts, reducing operator checks by **62%**.
- Improved model's inference rate by **30%** through image normalization, cropping ROI's, Color Space Optimisation, etc...
- Delivered **Python-based ETL workflows** generating CSV quality reports, enabling structured analysis of defect trends and inspection results for model fine-tuning.
- Supported integration into **Vision Autoflow** for real-time flagging of defective parts without disrupting other processes.

A-line Precision Tool Ltd.

Brampton, Ontario

CNC Machinist

Mar. - Aug. 2023

- Trained a team of 4 machinists on **GD&T**, machining nuances and optimised tooling, reducing shop downtime by **35%**.
- Documented process tweaks in procedures for intricate components for easy reproduction and reduced reworks or scraps.
- Oversaw shop-floor production of intricate components, reducing tool failures and unplanned downtime by **24%**.
- Guided operators on pre-inspection **quality checks**, identifying surface-finish issues, improving the shop efficiency by **44%**.

PROJECTS

Plant Disease Classifier | CNN and CV - based Classification System

Sep. - Dec. 2025

- Built a custom **CNN model** to classify 38 plant leaf diseases across 87,000+ images using **TensorFlow**, achieving **96%** accuracy on real world inference for a scalable agricultural and greenhouse monitoring.
- Administrated a **Python** pipeline with preprocessing procedures including pixel normalization, geometric augmentation for accurate predictions with least inference time and robustness against overfitting.
- Automated analysis workflows producing **F1 metrics, confusion matrices, and versioned CSV predictions**, enabling repeatable validation and production deployment readiness.
- Deployed the optimized CNN model on a **Raspberry Pi** for edge-based computer vision under compute constraints, achieving over **95%** inference accuracy in greenhouse-style deployment scenarios.

Servo Control Circuit Design | Pulse Width Modulation + Voltage Regulation

Sep. - Dec. 2025

- Designed a portable circuit using an **LM317T regulator** and **NE555**, delivering a stable **5.6V supply** from a 9V source.
- Calculated **PWM timing** to generate **0.5 - 2.5 ms pulses at 50 Hz** for the full operational range of a standard servo motor.
- Conducted **failure analysis** for resistors and capacitor networks, improving output stability by shifting from low-ohm to kilo-ohm resistors and reiterating designs to minimise jitter, improving motor consistency.

Stock Analyzer | Equity Analysis + Rest APIs

Mar. - Apr. 2024

- Built a **Python-based CLI** system integrating financial ratios, macro indicators, and sentiment analysis to generate **portfolio - specific trade advice**, built with a team of 4 for personal portfolio optimization
- Integrated **Rest APIs** e.g. YahooFinance, NewsAPI etc. for reliable live data ingestion which at deployment over market delivered \$1.2K realized and \$3K+ unrealized gains in 4 months
- Developed scalable, **Docker-containerized REST APIs** for financial data processing, enabling consistent backtesting and live deployments while reducing experimentation time by **90%**.

TECHNICAL SKILLS

Languages/Software Skills: Python, Java, C/C++, Embedded C, JavaScript, MySQL, MongoDB, Git, Docker, Linux

AI/ML: PyTorch, TensorFlow, TinyML, Scikit-learn, NumPy, OpenCV, Hugging face

Electronics: TinkerCAD, MicroCap, Soldering, PCB Design, Arduino, Raspberry Pi, ESP32

Mechanical: SolidWorks, AutoCAD, 3D Printing, Simulink, GD&T, Precision Machining, FEA, Quality Inspection

Certifications: CSWA – Certified SolidWorks Associate